Class 13:
Market Efficiency

- From ideas to model
- CAPM;
- Alpha and beta
- Luck vs Skill.
  - Caltech investors

CAPM (1) recap

- Someone has to own every asset.
  - Assets should earn the risk free return (plus some adjustment for risk)
- Representative agent =>The efficient portfolio is the value-weighted market.
  - Then every agent holds some combination of the market and some (short or long) riskless assets
- That means that the market portfolio best balances risk and return (its the tangent portfolio/Sharpe efficient portfolio
- Investor cares about Market portfolio risk (systematic risk)
  - Prices individual securities according to which they increase or decrease systematic risk

CAPM (1) recap

- Investor cares about Market portfolio risk (systematic risk)
  - Prices individual securities according to which they increase or decrease systematic risk.
  - Firms that add risk (their returns highly correlated to market) have high returns (because of the price of risk)
  - Firms that lower risk have low returns
- Need some way to compute expected return to every security
- Problem for today

Side Bar not in Lecture

- Security i pays no dividend and has an expected price p_i^t in t years.
- The expected return to buying that security and holding it for t periods.

$$r_i = \frac{p_i^t}{p_i^o} d^t$$

- If r_i is too low (high) to make demand =supply.
  - Need to adjust something.
  - Can't adjust p_i^t
  - have to adjust p_i^o its price now.
- Please check that $\uparrow r_i \Rightarrow \downarrow p_i^o$
- So what does CAPM tell us about r_i

The short answer

- Every security has a two-part return.
  - its return is the sum of the return to the risk-free asset and a risk premium.
- The risk premium of a security is simply its systematic risk (correlation with the market) times the price of systematic risk.
- All you need to know is the covariance of an asset with the market portfolio
  - Compute its means return $\mathrm{E}(r_i)$ $\sum_{\boldsymbol{r=1}}^{\boldsymbol{T}} \frac{\boldsymbol{r_{it}}}{\boldsymbol{T}}$
  - Compute is covariance β_{iM} ⇔
  - Cov $(r_i, p) = \sum_1^n (r_{\boldsymbol{it}} - \mathrm{E}(r_i))(r_{Mt} - \mathrm{E}(r_M))$
  - Its return $r_i = r_f + \beta_{iM}(r_p$ - $r_f) + \varepsilon$

Interpretation

- Its return $r_i = r_f + \beta_{iM}(r_M - r_f) + \varepsilon$
- $(r_M - r_f)$ is the price of systemic risk (the return to the market above the risk free rate)
- β_{iM} is how much asset i contributes to risk
  - $\beta_{iM} < 0$ asset i reduces risk =>its return is *less* than r_f
  - $\beta_{iM} > 0$ asset i increases risk =>its return is *higher* than r_f
  - Change in β_{iM} => changes in return => changes in prices.

How does this happen?

- The only way to make sure the equilibrium conditions hold is to adjust prices.
- So set prices so all the assets are in the portfolio
- The return to a CAPM portfolio is
- $r_M = (1 - \sum_{i=1}^{n} w_i)\, r_f + \sum_{i=1}^{n} w_i r_i$

  $= r_f + \sum_{i=1}^{n} w_i (r_i - r_f)$
- The marginal contribution to the portfolio return of asset i is it's return above the risk-free rate:

$$\frac{\partial r_M}{\partial w_i} = (r_i - r_f)$$

How does this happen (2) ?

- From previous
- $r_M = r_f + \sum_{i=1}^{n} w_i(r_f - r_i)$ and $\frac{\partial r_M}{\partial w_i} = (r_f - r_i)$
- The riskless asset has no variance (by definition) and no correlation with the other assets σ_f^2=0.
- The variance of the portfolio is the weighted variance of the market portfolio $\sigma_M^2 = \sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j \sigma_{ij}$
- σ_M^2 is just the market variance times the square of the market weight in the portfolio.

$$\sigma_M^2 = \left(\sum_{i=1}^{n} w_i\right)^2 \sigma_M^2$$

- The only terms that concern asset (i) are those in the ith row and ith column of the variance-covariance matrix.

- From returns
- $r_M = r_f + \sum_{i=1}^{n} w_i(r_f - r_i)$ and $\frac{\partial r_M}{\partial w_i} = (r_f - r_i)$

- The variance of the portfolio is the weighted variance of the risky portfolio $\sigma_M^2 = \left(\sum_{i=1}^{n} w_i\right)^2 \sigma_M^2$

- We can take the derivative to get the marginal contribution of asset i to portfolio variance

- $\boldsymbol{\frac{\partial \sigma_P^2}{\partial w_i} = 2w_i\sigma_i^2 + 2\sum_{j\neq i}^{n} w_j\sigma_{ij} \;=\; 2\sum_{i}^{n} w_j\sigma_{ij} = 2\sigma_{iM}}$

- That is just twice the covariance between the portfolio and the asset.

Volatility

- If the marginal contribution of asset i to portfolio variance is twice the covariance between the portfolio and the asset.

$$\frac{\partial \sigma_M^2}{\partial w_i} = 2\sigma_{iM}$$

- Then the marginal contribution of asset i to portfolio volatility is

$$\frac{\partial \sigma_M}{\partial w_i} = \frac{\sigma_{iM}}{\sigma_M}$$

Marginal risk return ratio

- Definition: the marginal risk return ratio (RRR) of asset i in a portfolio is the ratio of the marginal return to the marginal risk

$$RRR_i = \frac{\frac{\partial r_M}{\partial w_i}}{\frac{\partial \sigma_M}{\partial w_i}} = \frac{(r_i - r_f)}{\frac{\sigma_{iM}}{\sigma_M}}$$

- That tells us what happens to risk and return when we increase the weight of asset i.
- The bigger the number the more desirable it is to hold that asset.
- If these number vary across assets, then arbitrage possibilities.

Condition of equilibrium

- $RRR_i = RRR_j$
  - for all i,j—within the risky portfolio.
- $$RRR_i = \frac{(r_i - r_F)}{\sigma_{iM}/\sigma_M} = RRR_j$$

$$AND$$

$$RRR_i = RRR_M \Leftrightarrow \frac{(r_M - r_F)}{\sigma_M} = \frac{(r_i - r_F)}{\sigma_{iM}/\sigma_M}$$

Equilibrium

- But that implies
- $(r_i - r_F) = \frac{\sigma_{iM}}{\sigma_M^2}(r_M - r_F) = \beta_{iM}(r_M - r_F)$
- Where $\beta_{iM} = \frac{\sigma_{iM}}{\sigma_M^2}$
- So the Capital Asset Pricing Model implies
- $r_i = r_F + \beta_{iM}(r_M - r_F)$
- Where β_{iM} is the measure of asset i' s systematic risk in the market portfolio and$(r_M - r_F)$ is the price of systematic risk.

Security returns

- Security has a two-part return.
  - its return is the sum of the return to the risk-free asset and a risk premium.
- The risk premium of a security is simply its systematic risk (correlation with the market) time the price of systematic risk.
- Once you know the covariance of an asset with the market portfolio you can compute β_{iM} and then compute its return $r_i = r_f + \beta_M(r_M - r_f) + \varepsilon$
- The ε drops out because that risk averages out.
- Its price has to be of the form $p_i = \alpha + \beta(r_M - r_f)$

α adjusts for the intrinsic expected value of the firm

Two firms with the same variance but one has a higher expected return than the other…

$p_i - \alpha = \beta(r_M - r_f)$ is the price of risk

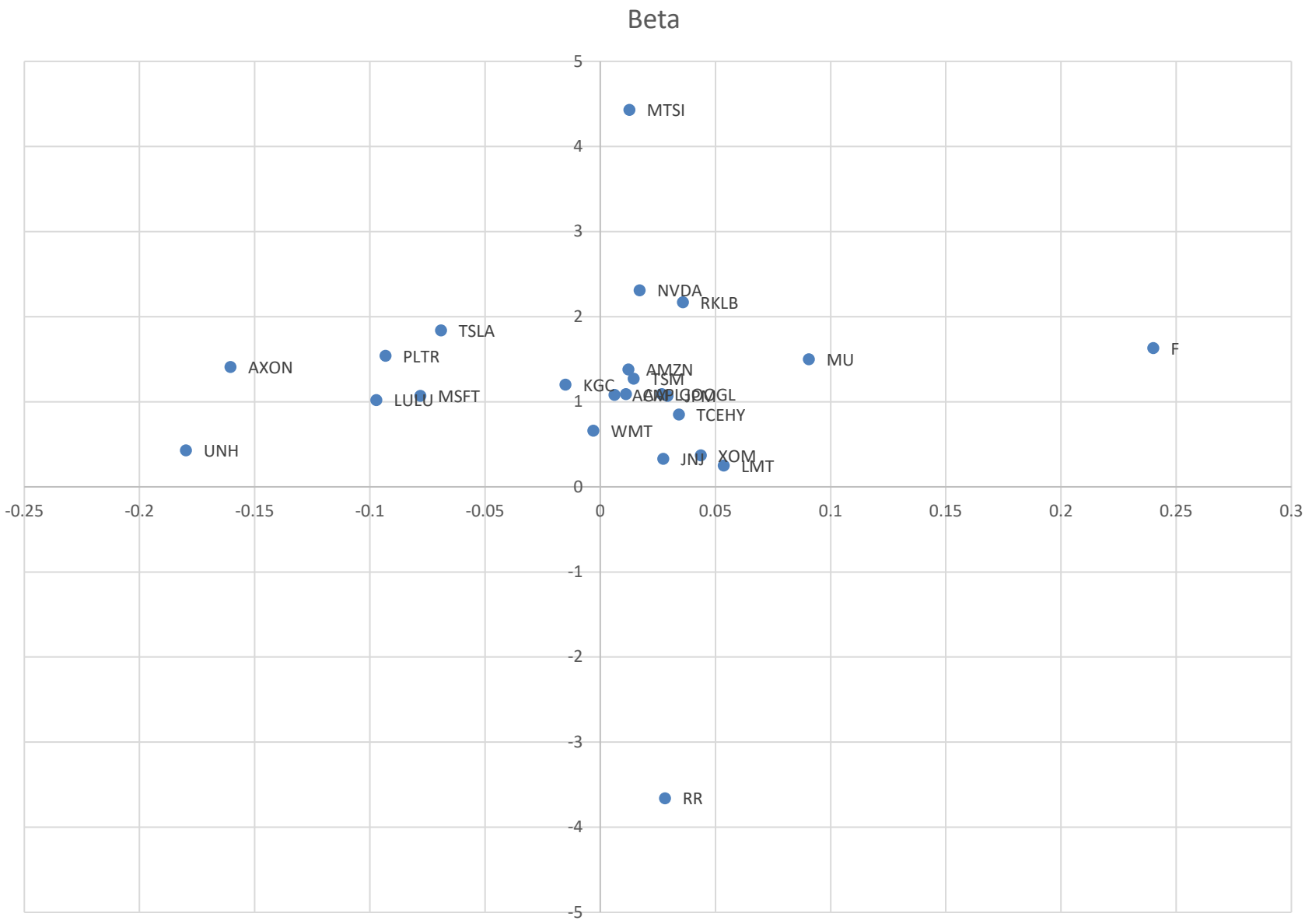
- More systematic risk (higher β_{iM})
  - > higher return because of the higher risk and you have to be to paid you more to hold it
  - > higher return => lower asset price
- Less systematic risk (smaller β_{iM})
  - > lower return because lower risk
  - > lower return => higher asset price.
- Security with a negative β_{iM} ($\sigma_{iM}<0$)
  - You are willing to ‘pay’ to hold it.
  - earn an expected return less than the riskless return.
- β_{iM}i s a risk factor (volatility that increases returns).

Practice

- Let the one year T-bill be the riskless asset, (that gives you r_F).
- Let the market be the S&P 500, (that gives you r_M, σ_M)
- Then given an asset you can estimate

$$(r_i - r_F) = \alpha_i + \beta_{iM}(r_M - r_F) + \varepsilon_i$$

- Assume (1) ε_i is IID normal with mean 0 and Cov(r_M, ε_i)=0.
  - you can estimate β_{iM} , σ_I, and α_i, Stdev(ε_i) by regression
- If CAPM holds α_i is zero going forward.
  - When it is not you face a pricing anomaly and you may try to take advantage of that, in particular, if it positive)
- β_{iM} gives you market sensitivity.
- ε_i is non-systematic risk (and if you are a long-term investor you should not care because it has mean zero). If you are a short-term investor you might worry if ε_i is large.


Beta
MTSI
NVDA
RKLB
TSLA
PLTR
AXON
MU
F
AMZN
TSM
KGC
LULU
MSFT
TCEHY
WMT
UNH
JNJ
XOM
LMT
RR
5
4
3
2
1
0
-1
-2
-3
-4
-5
-0.25
-0.2
-0.15
-0.1
-0.05
0
0.05
0.1
0.15
0.2
0.25
0.3

Market Efficiency

- How close are markets to CAPM?
- Close (very close) but not perfectly so
- If CAPM or another risk pricing model was perfect than estimate β_{iM}, σ_I, and α_i, Stdev(ε_i) by regression. Moreover, your estimate of the price of any given asset

$$p_i = \alpha + \beta(r_M - r_f) + \varepsilon$$

- It is the best you can do. There are no other variables that have statistical significance

Searching for risk factors

- Fama French
  - One is a size effect; small firms seem to earn higher returns than large firms, on average, even after controlling for beta or market sensitivity.
  - The other […] the ratio of a firm's accounting book value to its market value.
  - Momentum (firm's whose prices have gone up recently tend to go up again)

One factor many factors

- Since then
  - Harvey, Liu, and Zhu (2016) enumerate 316 "factor" candidates suggested in 313 papers […] Green, Hand, and Zhang (2013) identify 330 firm characteristics. K (Pukthuanthong et al. 2017)
- Some of this is 'academic' innovation
  - If a variable is highly correlated with size (age?) then its going to look like a factor, in particular in short run studies.

Implications

- All these additional factors are the result of a statistical analysis that says, the return of a set of securities is correlated with this variable (e.g. size)
  - On average small firms are cheaper relative to the variance of their returns relative to large firms.
  - So your returns are likely to be higher if you invest in smaller firms

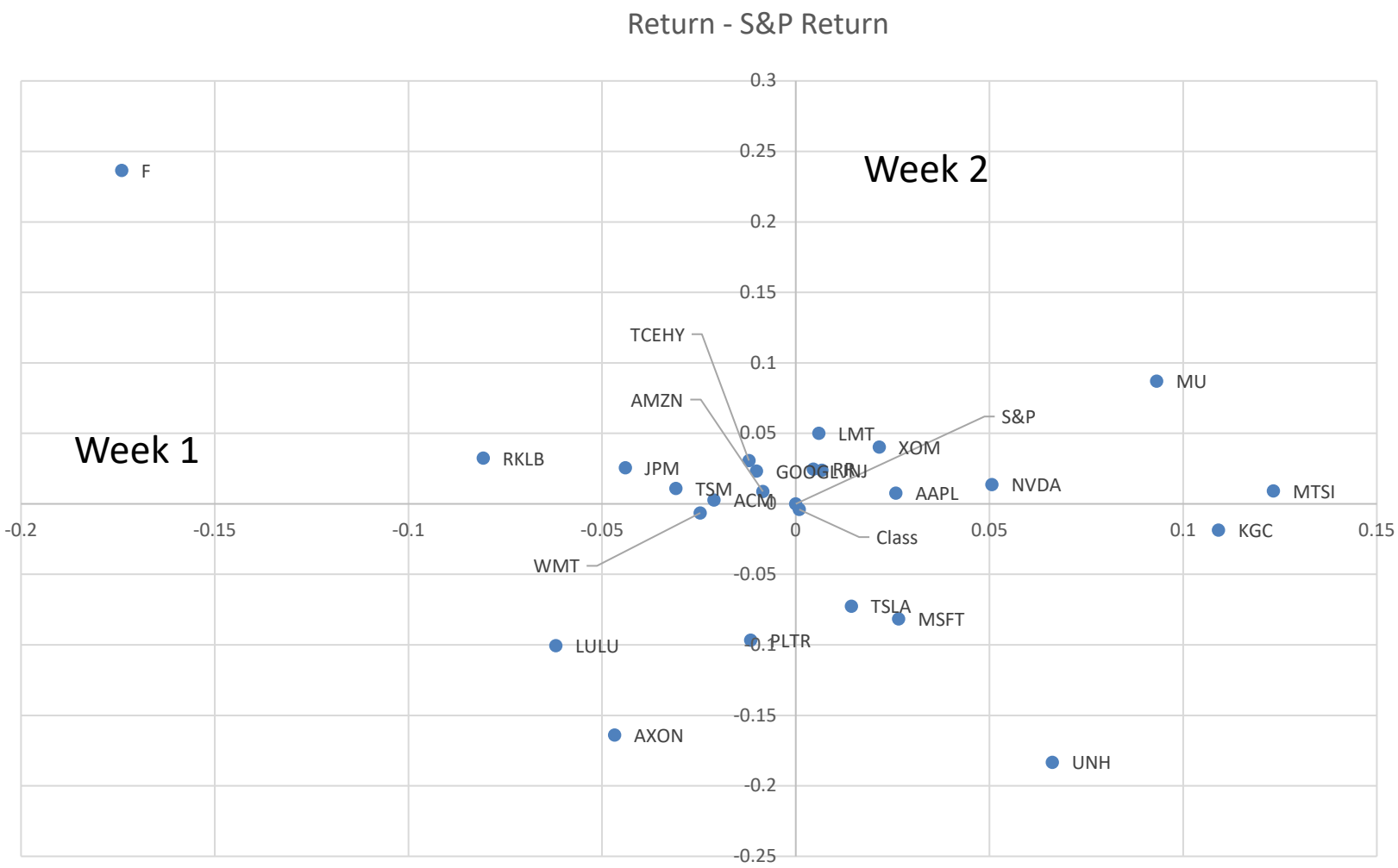
Back to investment funds.

- Index funds
  - Markets are efficient enough or
  - You believe in the one-factor model
- Passive funds (additional or smart β_i)
  - Multifactor market model that is better
  - Size, Momentum, GDP growth…
- Active funds (α)
  - Try to find funds that will have high return because …
  - Of something you have figured out
- Fee structure
  - Index is easy, when the set of companies does not change you do nothing except for inflow and outflows (on net these are small)..fees are not negligible
  - Passive is also relatively easy, except for tracking the β_i. But you do have to rebalance the portfolio regularly (towards mispriced firms)
  - Fees are now 0.25% or so
  - Active hardest because that is what everyone is looking for and prices are trying to get rid of this fees used to be 2% or so now about 1%.

Skill vs luck

- Lets start with 1,000 advisor who randomly chose stocks. On average they do no worse than the market.
- About half will have a portfolio that does worse than the market and half will do better after the first unit of time (year).
- Let them all make a second decision. The same will be true again, but randomness says
  - WW=0.25 WL+LW=0.5 LL=0.25. The number of very successful advisors is 250 after two years
- Now let's go out 10 months. 0.5^10=0.00977. There is one truly successful advisor.
- So success is necessary but nor sufficient to establish skill.

BEM 103 Stock Advisors



More details

- W1-W2 vs W2-W3 return correlation -0.29

Random			
		Week 1	
		Worse	Better
Week 2	Worse	6	6
	Better	6	6

Actual returns			
		Week 1	
		Worse	Better
Week 2	Worse	4	9
	Better	12	8

Recap

- Return to security = risk free return + price of risk
- Risk that matters is market risk. Volatility that is other than market risk is GOOD.
- If markets are efficient one factor model will explain everything that can be explained.
- If markets are efficient
  - You are wasting your time

Next time

- Conditional value at risk